NEUROAESTHETICS · MEASURED · MELANOPIC DOSE OF SCREEN PALETTES

Blue light filters don't work. *Palettes do.*

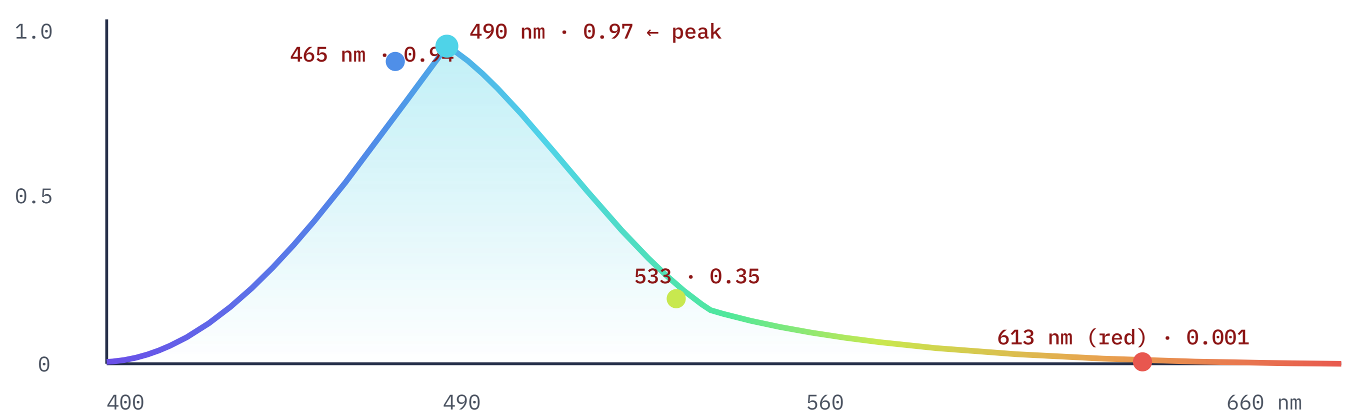
A visual neuroscientist recently argued that blue-light filters are near-useless, because **total luminance**, not blue hue, drives the melanopsin signal that suppresses melatonin. I built an instrument to measure it per pixel, and my own data agrees with him. But it also shows two things his piece doesn't: **"just go dark" isn't safe either**, and there is a fix a filter structurally cannot reach.

FIGURE 1 · THE RECEPTOR

MELANOPSIN SENSITIVITY, BY WAVELENGTH

The receptor peaks at cyan, not blue — and ignores red

Melanopsin (the ipRGC pigment that tells your brain it's daytime) peaks near 490 nm. A pure-red screen is almost invisible to it. This is why "blue" light is the wrong word — and why hue alone is a weak lever.



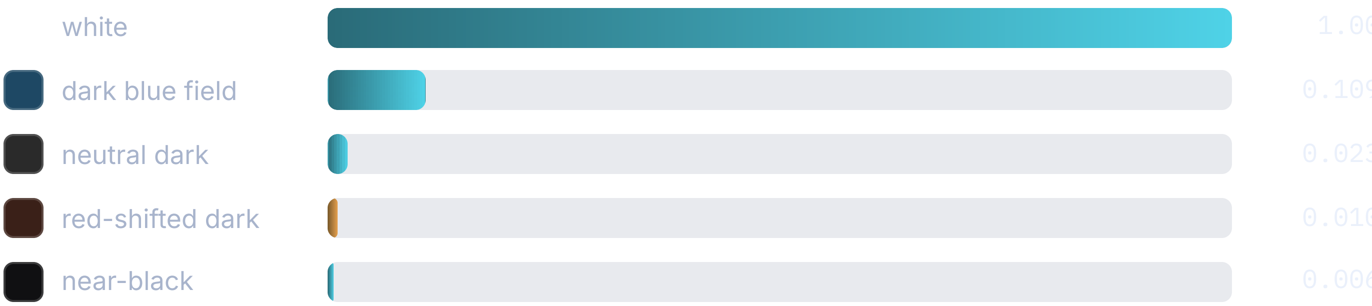
Generated from the Govardovskii et al. (2000) A1 pigment template, λ-max 480 nm (this model omits lens filtering, which shifts the published CIE S 026 peak to 490 nm). Values are the instrument's own output.

FIGURE 2 · THE WHOLE DOSE

MELANOPIC DOSE OF A FULL-SCREEN FIELD · WHITE = 1.00

Luminance dominates. He's right about that.

Dose = melanopic ratio × luminance. Drop the brightness and the dose collapses — every dark field is a fraction of white, regardless of hue. This is the filter's problem: it barely touches the number that matters.



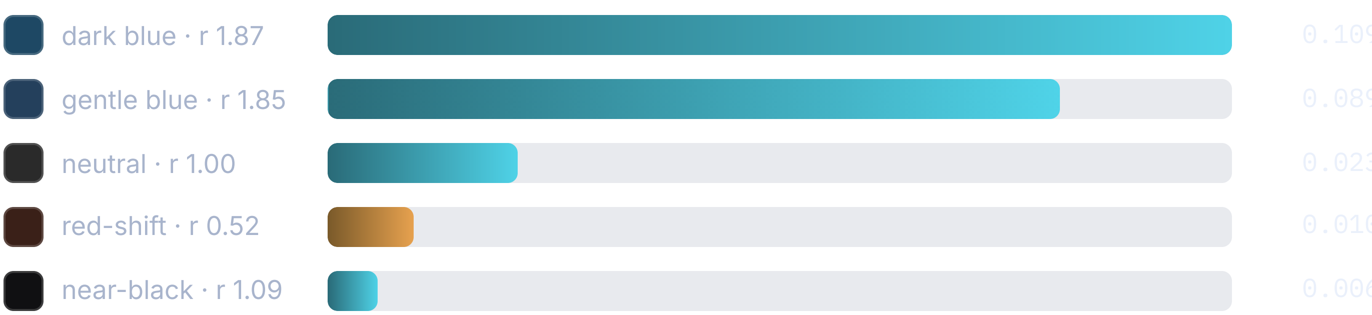
Relative melanopic dose, normalized so a D65-ish white = 1.00, on a modeled phone-OLED panel. Not lux — a ratio on one panel at one brightness.

FIGURE 3 · THE FINDING HIS PIECE DOESN'T HAVE

THE SAME DARK FIELDS, RESCALED — HUE SWINGS DOSE ~10×

"Just go dark" isn't safe. A gentle dark blue doses hardest.

Zoom into the darks and the hue reappears. My "calm" dark blue world carries a melanopic ratio of 1.87 — 87% more melanopic light per unit brightness than a neutral of the same lightness. Among fields that look equally dark, it doses ~4.7× a neutral grey and ~10× a red-shifted one.



Same numbers as Figure 2, scaled to the darkest set. r = melanopic-to-photopic ratio (white = 1.00); above 1.0 means the field is melanopically "hot" for its brightness. A dark theme built on blue can dose harder per lumen than white does.

FIGURE 4 · THE MOAT

WHY A FILTER CAN'T WIN, AND A PALETTE CAN

A filter dims the type with the field. A palette doesn't.

A screen filter multiplies every pixel by the same tint — glyph and field alike. So the only way it cuts real dose is by dimming everything, which kills the text. A palette treats them separately: push the *field* luminance down, hold the *ink* up to the legibility floor.

Screen filter

MULTIPLIES EVERY PIXEL EQUALLY

Streak · 12 days

`dose ↓ 21×  ·  but the type came with it`

Cut dose 21× and body text collapses to `APCA Lc 9.7` — invisible. Keep it readable and you cut dose just 1.5×. Nothing in between.

Palette

FIELD AND INK MOVED SEPARATELY

Streak · 12 days

`field ↓ hard  ·  ink held to the floor`

Field luminance stripped, type held bright: dose down to the 0.007–0.05 range and secondary text passes `APCA 8/8` — the first theme that's both dim and readable.

APCA = the perceptual contrast model used in the WCAG 3 drafts. The dusk theme this replaced passed 2/8; the palette-restructured one passes 8/8 while cutting measured melanopic dose.

> *The reason a filter fails isn't that it's too weak. It's that it can't tell a* **glyph** *from a* **field** *— so it has to choose between your sleep and your eyes. A palette refuses the choice.*

WHERE I AGREE, AND WHERE I GO FURTHER

Agree

Filters are near-useless and luminance dominates hue. My instrument confirms it independently: hue-shift alone buys only a `1.5×` dose cut, because dose is mostly a brightness story.

Further

"Go dark" isn't the whole answer. A dark blue field can be melanopically hotter than white per lumen (`r 1.87`), and the real lever — separating ink from field — is one a filter structurally cannot pull.

> **What this is and isn't.** These are relative ratios on one modeled OLED panel at one brightness — not lux, and not a clinical melatonin measurement (the consensus thresholds, Brown et al. 2022, are in mEDI units this model can't produce). Lens filtering is omitted, so the modeled peak sits at 480 nm vs the CIE-published

> 490. The absolute numbers are indicative; the **ratios between palettes on the same model** are the point, and the blue-vs-red ratio spans two orders of magnitude, which no plausible modeling error reverses.

Measured and written by **Marion Moranetz**. Instrument (melanopsin template, per-pixel dose, the APCA + dose gate) open on request.

Written in reply to Patrick Mineault's "Blue light filters don't work" (neuroai.science) and his colour-perception project ismy.blue — this extends his luminance argument with per-pixel measurement and a design method. Govardovskii et al. (2000); CIE S 026:2018; Brown et al. (2022); APCA / WCAG 3.